

Infiltration and solute transport under a seasonal wetland: bromide tracer experiments in Saskatoon, Canada

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Abstract:

In the northern glaciated plain of North America, the duration of surface water in seasonal wetlands is strongly influenced by the rate of infiltration and evaporation. Infiltration also plays important roles in nutrient exchange at the sediment–water interface and groundwater recharge under wetlands. A whole-wetland bromide tracer experiment was conducted in Saskatchewan, Canada to evaluate infiltration and solute transport processes. Bromide concentrations of surface water, groundwater, sediment pore water and plant tissues were monitored as the pond water-level gradually dropped until there was no surface water. Hydraulic head gradients showed strong lateral flow from under the wetland to the treed riparian zone during the growing season. The bromide mass balance analysis showed that in early spring, almost 50% of water loss from the wetland was by infiltration, and it increased to about 70% in summer as plants in and around the wetland started to transpire more actively. The infiltration contributed to recharging the shallow, local groundwater under the wetland, but much of it was taken up by trees without recharging the deeper groundwater system. Emergent plants growing in the wetlands incorporated some bromide, but overall uptake of bromide by vegetation was less than 10% of the amount initially released. After one summer, most of the subsurface bromide was found within 40–80 cm of the soil surface. However, some bromide penetrated as deep as 2–3 m, presumably owing to preferential flow pathways provided by root holes or fractures. Copyright © 2004 Crown in the Right of Canada. Published by John Wiley & Sons, Ltd.

KEY WORDS wetland; riparian; bromide; groundwater recharge; evapotranspiration

INTRODUCTION

The northern glaciated plain of North America has numerous topographic depressions. Prairie wetlands forming in these depressions do not have permanent surface inflow or outflow, and are regarded as hydrologically closed basins. The water level in prairie wetlands is generally highest in early spring as a result of snowmelt runoff, and gradually declines in summer primarily owing to evaporation and infiltration exceeding the inputs of rain and occasional storm runoff (Winter, 1989). Many prairie wetlands hold surface water only for a few weeks to months. The duration of surface water is a critical habitat parameter for waterfowl and other species dependent on water (Swanson and Duebbert, 1989).

Aquatic plants growing within and around surface water bodies probably have a major effect on infiltration rates. For example, Millar (1971) showed that infiltration rates under prairie wetlands increased as the ratio of the wet perimeter length to the wet area increased, implying significant shoreline ‘loss’ owing to evapotranspiration by riparian plants. Winter and Rosenberry (1995) and Rosenberry and Winter (1997) showed that the groundwater flow direction frequently reversed between prairie wetlands and the riparian zone. The flow direction was towards the riparian zone during dry periods, and it was towards the wetland during wet periods when the water table under the riparian zone rose above the wetland water level. Commonly

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observed diurnal fluctuations of pond water-level (Rosenberry and Winter, 1997; Hayashi *et al.*, 1998a) are related to the diurnal cycle of evapotranspiration. Infiltration under prairie wetlands, by definition, is the recharge of shallow, local groundwater systems connected to the wetlands. However, much of the infiltration and local groundwater recharge is only temporary, and eventually is consumed by evapotranspiration without recharging larger and deeper groundwater systems (van der Kamp and Hayashi, 1998).

Infiltrating water transports dissolved material and colloid-size particles. The transport processes are important for nutrients cycling within wetlands, and also for potential contamination of shallow groundwater by agricultural pesticides transported into wetlands by runoff (Donald *et al.*, 1999). Sediments underlying the wetlands have clay-rich texture, but probably have numerous macropores composed of root holes and desiccation fractures (Winter and Rosenberry, 1995), which may provide preferential transport pathways (Kamau *et al.*, 1994). Therefore, a commonly used 'piston flow' model of solute transport, where a uniform front moves downward with infiltration, may not be applicable, but this process has not been well documented for prairie wetlands.

Previous hydrological studies of prairie wetlands mostly relied on hydraulic measurements and water-balance calculations to understand the recharge and discharge of groundwater (Meyboom, 1966; Zebarth *et al.*, 1989; Woo and Rowsell, 1993; Winter and Rosenberry, 1995; Hayashi *et al.*, 1998a). This approach provides useful information on the spatially averaged water flux under an entire wetland or the distribution of hydraulic gradients within the wetland. However, owing to a large degree of uncertainty in hydraulic conductivity, it is difficult to quantify the flux distribution within the wetland. Tracer experiments provide direct information on the distribution of water and solute flux. This paper presents the results of a whole-wetland experiment for which sodium bromide was used as a tracer.

The objectives of the study are:

1. compare the relative importance of open-water evaporation and infiltration;
2. delineate the spatial distribution of infiltration and bromide transport flux;
3. examine the validity of viewing solute transport as a piston flow;
4. estimate the amount of bromide uptake by the emergent vegetation.

Bromide is commonly assumed to be conservative, but the literature suggests that plants may take up a significant amount of bromide (Flury and Papritz, 1993). Therefore, it is important to evaluate the relative amount of the bromide uptake by plants in comparison to the overall mass in the system.

STUDY SITE

The experiment was conducted at wetland 109 located in the St Denis National Wildlife Area (106°06'W, 52°02'N), approximately 40 km east of Saskatoon, Saskatchewan, Canada. Wetland 109 and other, similar wetlands are situated in a cultivated field of about 1 km² area that lies on a regional high about 10–15 m above the floor of a surrounding valley. The area has a hummocky topography and is underlain by a clayey glacial till. The till is weathered to a depth of approximately 5 m, as indicated by an olive-brown colour, and is underlain by grey unweathered till. Thin, discontinuous sand lenses are scattered throughout. A continuous, 0.3-m-thick clay layer occurs at about 8 m depth, and a 1.5-m-thick sand aquifer lies at a depth of approximately 25 m. A number of piezometers with depths ranging between 1.4 and 22 m were installed by Miller *et al.* (1985) and Hayashi *et al.* (1998a) in wetland 109 and the surrounding area (Figure 1). These are called old piezometers. The hydraulic conductivity determined by slug tests on old piezometers is in the order of 10⁻⁸ to 10⁻⁶ m/s for the weathered till, and it decreases to 10⁻¹¹ to 10⁻⁹ m/s in the unweathered till as the occurrence of conductive fractures decreases (Hayashi *et al.*, 1998a). The main focus of the present study is on the top 3 m of soil, in which bromide concentration and hydraulic heads were monitored. The soils underlying the wetland are classified as a Humic Luvic Gleysol (Miller *et al.*, 1985).

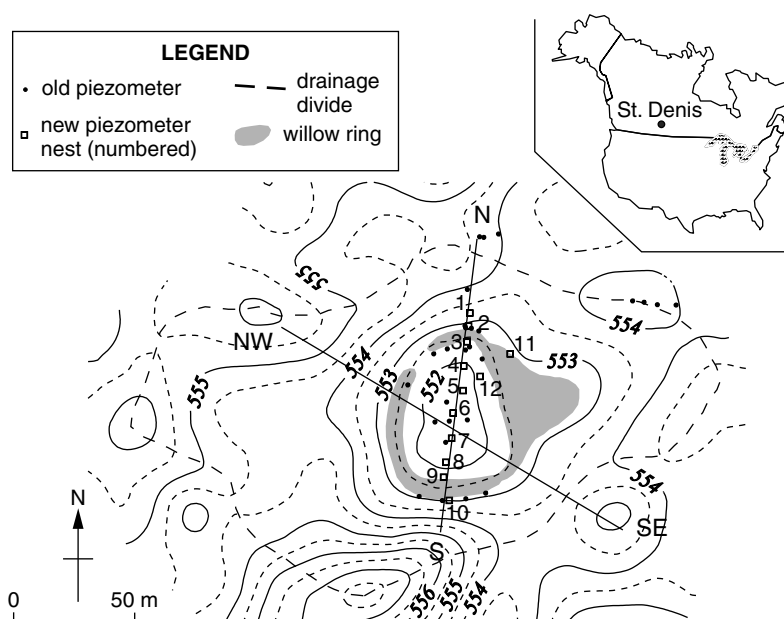


Figure 1. Site map of wetland 109 showing the location of piezometers. The elevation is in meters above mean sea level. The inset is a map of Canada and USA with the location of St Denis National Wildlife Area

Crops on the surrounding field consisted of spring wheat in 1999, and peas and lentils in 2000. Swamp smartweed (*Polygonum coccineum* Muhl.), cow parsnip (*Heracleum lanatum* Michx.), sedge (*Carex* sp.) and grasses grew in and around the central pond over both summers, and grew to heights of over 1 m. Willow (*Salix* sp.), trembling aspen (*Populus tremuloides*) and balsam poplar (*Populus balsamifera*) grow along the wetland margin forming a 'willow ring'. The willow ring is mostly about 10 m wide or less, but an area of mainly balsam poplar on the east side is approximately 30 m across. Willows on the north and west sides were not much higher than about 3 m, whereas aspen and poplar ranged in height from 5 to 8 m.

Monthly mean temperatures in Saskatoon range from -19°C in January to 18°C in July (MSC, 2001). The mean annual precipitation is 360 mm, with about 280 mm occurring as rainfall in April–October. Annual lake evaporation in this area is approximately 700 mm (Morton, 1983). Interannual variations in climatic factors, precipitation in particular, are the major controls over vegetation and hydrological conditions in the wetland (Winter and Rosenberry, 1998). Over the past 30–35 years, maximum yearly pond-water depths have ranged between 0 and 1.2 m (van der Kamp *et al.*, 2003). In 1999, the spring pond depth reached 0.52 m, but the pond was only 0.37 m deep in spring 2000.

Several different terms will be used here to refer to different areas in and around wetland 109. The term 'pond' refers to the standing water area in the depression, whereas 'wetland' refers to the depression itself and the surrounding willow ring. The term 'watershed' will be used for the entire area within the drainage divide (Figure 1), which has an approximate area of 20 000 m².

METHODS

Water level and precipitation measurement

In October 1998, 12 new piezometer nests were installed (Figure 1). The water table was located 1.5–2.0 m below the ground surface at the time of installation. Table I lists the dimension of piezometer screen or sand pack. Each nest contained at least three 'steel' piezometers. Nests located inside the willow ring each

Table I. Piezometer types, casing inner diameter, the presence of sand pack, and screen or sand pack length and diameter

Type	Depth range (m)	Casing inner diameter (cm)	Sandpack	Screen/sand-pack length (cm)	Screen/sand-pack diameter (cm)
Mini	0.2–0.6	0.43	No	5	0.8
Steel	0.8–2.1	0.92	Yes	10	1.4
1/2-inch	3.0	1.3	Yes	20–25	6.4
Old	1.4–7.6	3.2	Yes	70–180	15

included a bundle of three minipiezometers, and odd-numbered nests along each transect included a ‘half-inch’ piezometer. Table I also lists the dimension of old piezometers installed by Miller *et al.* (1985) and Hayashi *et al.* (1998a) and used in this study.

Minipiezometers consisted of segments of 0.43-cm inside diameter (ID) polyethylene tubing bundled and tied to an aluminum rod. Each minipiezometer bundle was placed in a hole made with a 2.5-cm diameter soil sampler, and back-filled with soil to the ground surface. The steel piezometers consisted of 0.92-cm ID stainless steel tubes that were pushed and hammered to their respective depths. A point-tipped brass insert was placed at the leading end of the piezometer tube to prevent soil from entering. The shaft of the insert was cylindrical with a diameter of 0.7 cm and the base diameter of the cone-shaped tip was 1.7 cm, slightly larger than the outside diameter of the tube (1.4 cm). When the tube was inserted to the desired depth, a small quantity of sand was poured into the open end, and the tube then was pulled up 10 cm. This left a sand pack approximately 10 cm long underneath the bottom end, and above the brass insert. The half-inch piezometers were constructed from 1.3-cm ID polyethylene tubes with slotted polyethylene screens fitted over one end. These tubes were placed in 6.4-cm diameter hand-augured holes. The screen was surrounded by a sand pack approximately 20 cm long, and the rest of the hole was filled to the top by bentonite pellets and chips. In comparison with a previous study of wetland geochemistry by Hunt *et al.* (1997), our mini- and steel piezometers have the ‘intermediate’ sampling scale, and our half-inch and old piezometers have the ‘large’ sampling scale.

Bentonite products used in this study did not contain detectable levels of bromide when they were tested in the laboratory using a method similar to Remenda and van der Kamp (1997). The elevations of all piezometer casing tops were surveyed in the summers of 1999 and 2000. Surveys were conducted in both years to prevent measurement errors as a result of frost heaving during the spring (Conly and van der Kamp, 2001).

Slug tests (Hvorslev, 1951) were performed on most of the newly installed piezometers to measure their basic time-lags. Most of the piezometers were developed before slug tests by purging water several times. The response of piezometers installed in clayey glacial tills, such as this site, is essentially governed by the flow in the low-permeability till, not by the flow within sand pack or screen. Therefore, the development of these piezometers is not as critical as piezometers installed in high-permeability materials. Hydraulic conductivity was estimated from the basic time lag and the dimension of sand packs or, in the case of minipiezometers, screens (Table I). Water levels of the pond were measured half-hourly using a vibrating-wire pressure transducer (Geokon, 4500ALV) placed at the bottom of a 3.8-cm diameter stilling well located near the pond centre. Pond water volume, V (m^3), and pond area, A (m^2), were calculated from the depth of water, h (m), at the deepest part of the pond, using the volume–area–depth functions (Hayashi and van der Kamp, 2000)

$$A = 3180 h^{1.24} \quad (1)$$

$$V = 1420 h^{2.24} \quad (2)$$

Precipitation was measured with a tipping-bucket rain gauge, located within the watershed, between 7 April and 3 November 1999, and between 16 March and 28 July 2000. Winter precipitation was estimated using data from Saskatoon airport (MSC, 2001).

Introduction of tracer and water sampling

The tracer was applied to the central pond on 28 April 1999, when the depth and volume of pond water were 0.45 m and 240 m³, respectively. Portions of approximately 4 kg of technical grade sodium bromide (Van Waters and Rogers, Ltd) were each mixed with pond water in 20-L polyethylene containers. To achieve an even application of tracer, the containers were emptied through a spigot from the back of a small boat, as it was being paddled around the pond. This was repeated 10 times, thus introducing 40 ± 0.5 kg of sodium bromide (24 kg of bromide ions) to the pond and increasing the bromide concentration of pond water to almost 100 mg/L. Bromide was chosen as the tracer because it is considered to be relatively conservative, and because background concentration is very low (Flury and Papritz, 1993).

Surface water was sampled at five locations in the pond; at the centre and at each of the north, south, east and west corners, about 2 m from the water edge. Water collected within an hour of tracer application ranged in bromide concentration from 50 to 150 mg/L. The samples taken 2 days later all had similar concentrations with an average of 98 mg/L, which was close to the expected concentration of 100 mg/L, showing that the pond was well-mixed. A small boat was used to access each sampling location to minimize disturbance of the bottom sediments. Sampling was done at these locations until July 1999, after which the pond area was too small to make it practical. Sampling was done only at the pond centre after this. At each location, a 60-mL polyethylene bottle was submerged, filled to the top, capped and labelled. Pond water was sampled on a weekly basis.

Piezometer water was sampled by suction from the bottom through a polyethylene tube using a syringe. Minipiezometers were sampled in a similar fashion, with the syringe being connected directly to each piezometer tube. Piezometers were completely emptied at each sampling event. The low hydraulic conductivity of the till and the slow response of many of the piezometers made purging of piezometers impractical. Therefore, each sample represented the water that entered the piezometer between the current and previous sampling event. Piezometers with screens or sand packs above the water table were sampled on a weekly to monthly basis in spring and summer of 1999 and 2000. All water samples were filtered through 0.45- μ m cellulose nitrate membranes, stored at 4°C for several days to weeks, and analysed for bromide by ion chromatography (IC).

Soil sampling and pore-water extraction

Soil samples for pore-water extraction were collected from the vadose zone in October 1999, May 2000 and July 2000 using 6.4-cm diameter hand augers at 10-cm depth interval. Samples were disturbed in that they were cut and twisted by auger blades, but most were able to maintain their texture and cohesiveness. Augered holes were located at the mid-points between successive pairs of new piezometer nests. All of the holes from which samples were obtained were later filled to the surface with bentonite chips. Samples were sealed in plastic bags, and stored at 4°C for several days to weeks before pore water was extracted for analysis. Most of the soil samples consisted of naturally consolidated glacial till containing pebbles, from which very little water could be squeezed by compression. Soil solution samplers are not effective in unsaturated till matrix that typically has hydraulic conductivity less than 10^{-9} m/s (Hayashi *et al.*, 1997). Therefore, aqueous extraction is the only practical method for extracting bromide from the unsaturated soil. Approximately 50 g of each sample was oven-dried for 24 h at 105°C and weighed to determine gravimetric water content. Another 100 g of wet sample was placed in a 250-mL polyethylene bottle, to which deionized water was added. The dilution factor was determined from the amount of water added and the measured gravimetric water content. Sample-water mixtures were placed on a mechanical wrist-action shaker and shaken vigorously for 4 h. Shaken samples were centrifuged at 7000 rpm for 0.5 h, and supernatant was collected in small sample vials. The procedure

for pore-water extraction was based on a method described by Rhoades (1982). Supernatant was filtered through 0.45- μm cellulose nitrate membranes and analysed for bromide by IC.

Twenty holes, 0.9 to 1.5 m deep, were drilled along transects N–S and NW–SE (see Figure 1) in the autumn of 2000 using a truck-mounted auger and a hand auger to describe soil-horizons (D. Cerkowniak, personal communication). Undisturbed soil-core samples for porosity and bulk density measurements were collected from locations shown in Figure 2. The cores were sampled using 5-cm diameter steel core tubes at depth intervals of 0–0.15, 0.15–0.3, 0.3–0.6, and 0.6–0.9 m at all holes, and deeper intervals at some holes. Soil dry bulk density was determined from weight and volume measurements. Porosity was estimated assuming an average particle density of 2650 kg/m³ representing aluminosilicate minerals (Hillel, 1998, p. 315). This particle density may slightly underestimate that of carbonate-rich soil and overestimate organic-rich soil, but is considered appropriate for the purpose of this study.

Larger undisturbed cores were collected in July 2001 for the laboratory measurement of hydraulic conductivity, and also for bulk density and porosity. The cores were obtained from various depths up to 2 m at five locations within the wetland. At each location a 10-cm diameter hole was drilled by a truck-mounted auger to the desired depth, and intact core samples were obtained using 7.6-cm diameter Shelby tubes pushed 20 cm into soil at the desired depth. The inside walls of tubes were sprayed with light lubricating oil to minimize the friction-induced compaction of soil. Falling-head tests were conducted on 20-cm-long soil cores to determine saturated vertical hydraulic conductivity, using a deaerated 0.005-M calcium sulphate solution (Klute and Dirksen, 1986).

Vegetation sampling and chemical analysis

Different types of vegetation were present in different areas of wetland 109 and its watershed in the summer of 1999 (Figure 2). These different areas defined vegetation ‘zones’ that provided the basis for the choice of locations sampled in early September 1999. Vegetation was sampled inside a 1-m² wooden quadrat placed at the approximate mid-point of each vegetation zone along north–south and east–west transects. All emergent plant material within the frame area was cut using garden shears and bagged. Only above-ground portions of vegetation were obtained, as root systems typically make about 10% or less of the dry mass of such plants, and most solutes that are incorporated into plants by root uptake end up in leaves and stems (Karcher, 1995, p. 144).

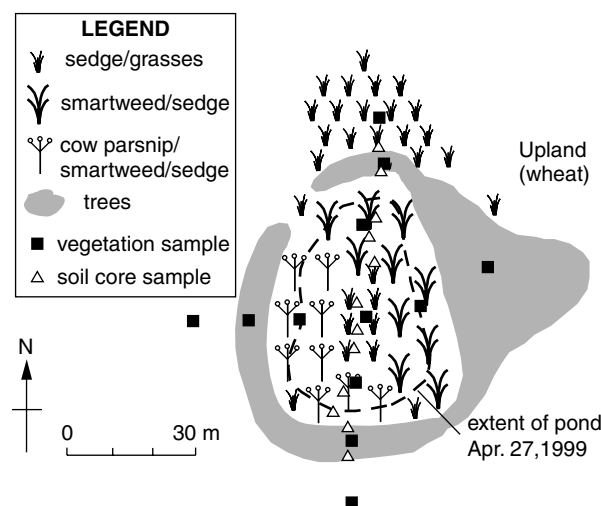


Figure 2. Map of vegetation zones and sampling locations

Total biomass of trees was estimated in two steps. First, the volume of a tree trunk was calculated by taking it to be a cylinder (Chapman, 1976, p. 174). Next, average-sized branches were taken from each tree within the sampling area, and each of their masses was multiplied by the number of branches present. The mass of the trunk was roughly estimated from a density measurement of a small, cylindrical segment of the branch sample.

The fresh plant material was stored for several weeks in a freezer at -10°C . The samples were subsequently thawed, weighed, rinsed with deionized water and then oven-dried at 70°C for 24 h and reweighed. The reason for drying at a relatively low temperature was to avoid any volatilization of chemical species present in the plant material (Walinga *et al.*, 1995). The dried material was broken down by hand and passed through a 1-cm mesh. Each sample was then split to obtain small subsamples. These were processed through a Wiley mill with a 1-mm mesh, to form fine, needle-like fragments (Walinga *et al.*, 1995). Other dry subsamples were weighed and oven-dried again at 105°C to determine water content of the plant tissues. One gram of each milled sample was placed in a 100-mL flask with 50 mL of deionized water, shaken for 0.5 h, and passed twice through a filter paper. Filtrate was collected in small vials, and was filtered again through a $0.45\text{-}\mu\text{m}$ membrane in preparation for IC analysis.

RESULTS

Surface water

In 1999, the water depth in wetland 109 reached a peak of 52 cm after the completion of snowmelt in mid-April (Figure 3). The water level in piezometers under the pond responded to flooding somewhat slower (Figure 3) presumably because the frozen sediments did not allow rapid infiltration (Hayashi *et al.*, 2003). The pond level then began to drop steadily, and the bromide tracer was introduced on 28 April 1999, when the pond depth was about 45 cm. Before introduction of the tracer, pond bromide concentrations were below the IC detection limit (0.1 mg/L). The bromide concentration rose to 98 mg/L upon the tracer release, and gradually declined as a result of dilution by precipitation (Figure 3). Runoff over the cultivated uplands is very rare in summer (Hayashi *et al.*, 1998a), but significant runoff can occur within the wetland where the

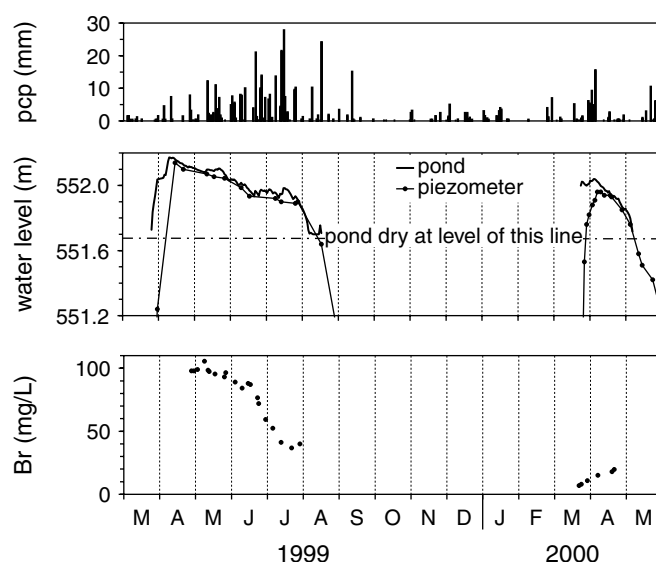


Figure 3. (a) Daily precipitation. (b) Pond and piezometer water level. (c) Bromide concentration in the pond. The piezometer is in nest 5 (see Figure 1) and has a screen at 3 m

top of the capillary fringe is close to the ground surface (Gerla, 1992). The amount of runoff, indicated by the difference between precipitation and pond water-level rise, was negligible for rainfall events less than 10 mm, but there were several heavy rain events, during which pond level rose as much as 10 mm more than the amount of precipitation. Pond level declined quickly after conditions became drier in late July, and the pond was dry by 9 August, except for some small puddles at the centre that were sustained through mid-August by a few more heavy rain events. Piezometer water level consistently stayed a few centimetres below the pond level (Figure 3), indicating the downward flow of water.

A combination of thin winter snow cover and unusually warm temperatures in early March resulted in a relatively small amount of spring runoff in 2000 (van der Kamp *et al.*, 2003). The pond reached a maximum depth of about 37 cm in late March, and had become completely dry by early May (Figure 3). The piezometer water level lagged about 2 weeks behind the pond water level and stayed consistently lower than the pond level.

Even though the new pond water in 2000 came from snowmelt that should contain no bromide, the tracer did appear in a measurable quantity in the pond. The bromide concentration increased from 7 mg/L on 22 March to 20 mg/L on 21 April with little rainfall during this period. Most of this bromide was probably incorporated from the sediments into pond water by exchange processes at the water–sediment interface. The mass of bromide in pond water reached the maximum of 2.1 kg on 7 April and started decreasing owing to infiltration loss.

Soil physical properties

Figure 4 shows the distribution of soil horizons along N–S and NW–SE transects in Figure 1. The A-horizon consists of dark brown, granular soil that is highly porous and rich in organic material. Underlying this is the B-horizon, defined by the absence of carbonate, which is made up of dark greyish-brown clay containing plentiful root fragments as well as reddish-brown mottling at the top and bottom. The B-horizon extends to about 1 m depth near the centre to almost 1.5 m beneath the edge of the wetland. Below this is greyish-brown to yellowish-brown silty clay till which makes up the C-horizon. At these depths, abundant

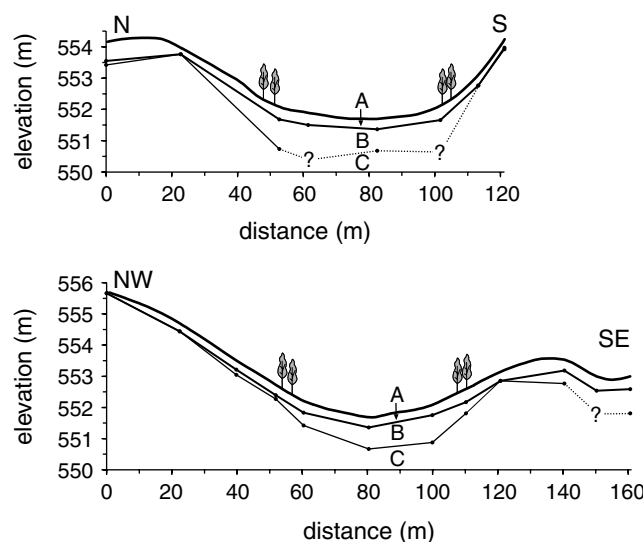


Figure 4. Simplified soil profile cross-sections along transects N–S and NW–SE (see Figure 1 for locations). Solid circles indicate the boundary between soil horizons. The ‘?’ symbols indicate the bottom of the holes, which were still in B horizons, and dashed lines indicate shallowest possible boundaries between B and C horizons

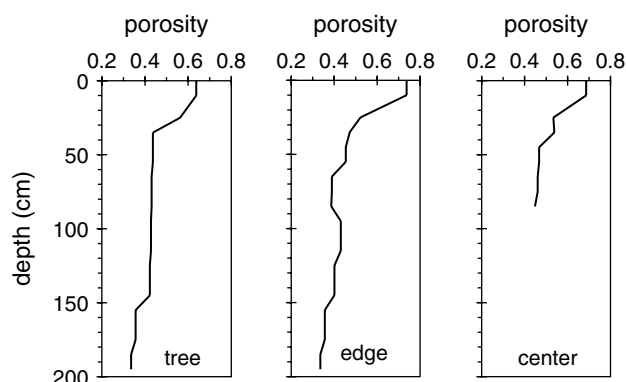


Figure 5. Average porosity profiles for the pond centre, pond edge and willow ring (tree) zones

reddish-mottling, sand lenses, pebbles and some decayed root fragments are encountered. The C-horizon directly underlies the A-horizon underneath the upland field outside the willow ring.

Figure 5 shows porosity profiles for the willow ring (indicated as 'tree'), pond edge and pond centre areas. The porosity profiles represent the average of all samples collected in the respective zones (see Figure 2). Porosity generally decreased from between 60 and 75% in the A-horizon to between 35 and 50% in the B-horizon, and to about 30% in the C-horizon.

Hydraulic conductivity measured by slug tests in mini- and steel piezometers had a large variability, whereas half-inch and old piezometers measured more consistent and generally higher values (Figure 6). The latter piezometers had much larger sand packs than the former (Table I), which gave the latter higher chances of intersecting conductive fractures and macropores resulting in generally higher conductivities (Keller *et al.*, 1988). Hydraulic conductivities determined from falling-head tests of soil core samples were higher in the top 1 m or so than the deeper zones (Figure 6). Slug tests are normally considered more representative of the field condition than falling-head tests on soil cores because the former integrates both vertical and horizontal conductivities whereas the latter measures vertical conductivity alone. However, in this study falling-head tests had a much larger sampling volume (7.6 cm in diameter by 20 cm in length) than mini- and steel piezometers and, hence, were regarded as a more reliable indicator of bulk hydraulic conductivity. Visual observation of samples revealed that the higher-conductivity samples had higher sand content in some cases, and abundant

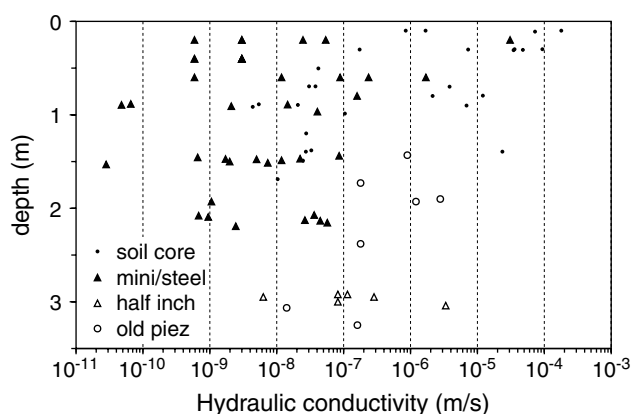


Figure 6. Vertical distribution of saturated hydraulic conductivity under wetland 109 measured with slug tests on piezometers and falling-head tests on core samples

roots and root holes in others. The lower conductivity samples were generally massive and clayey. It is most likely that high variability of conductivity in the top metre reflects the difference in soil texture as well as the presence of macropores.

Groundwater

Groundwater flow immediately beneath the pond was downward and laterally divergent towards the pond edges through most of the spring and summer of 1999, except for a 'flow-through' condition (LaBaugh *et al.*, 1987), which may have existed for a brief period in early spring when the water table and hydraulic heads were relatively high to the south of the pond (Figure 7a). In summer, as wetland vegetation began to transpire more actively, a 'water table trough' (Rosenberry and Winter, 1997) and hydraulic head lows occurred beneath

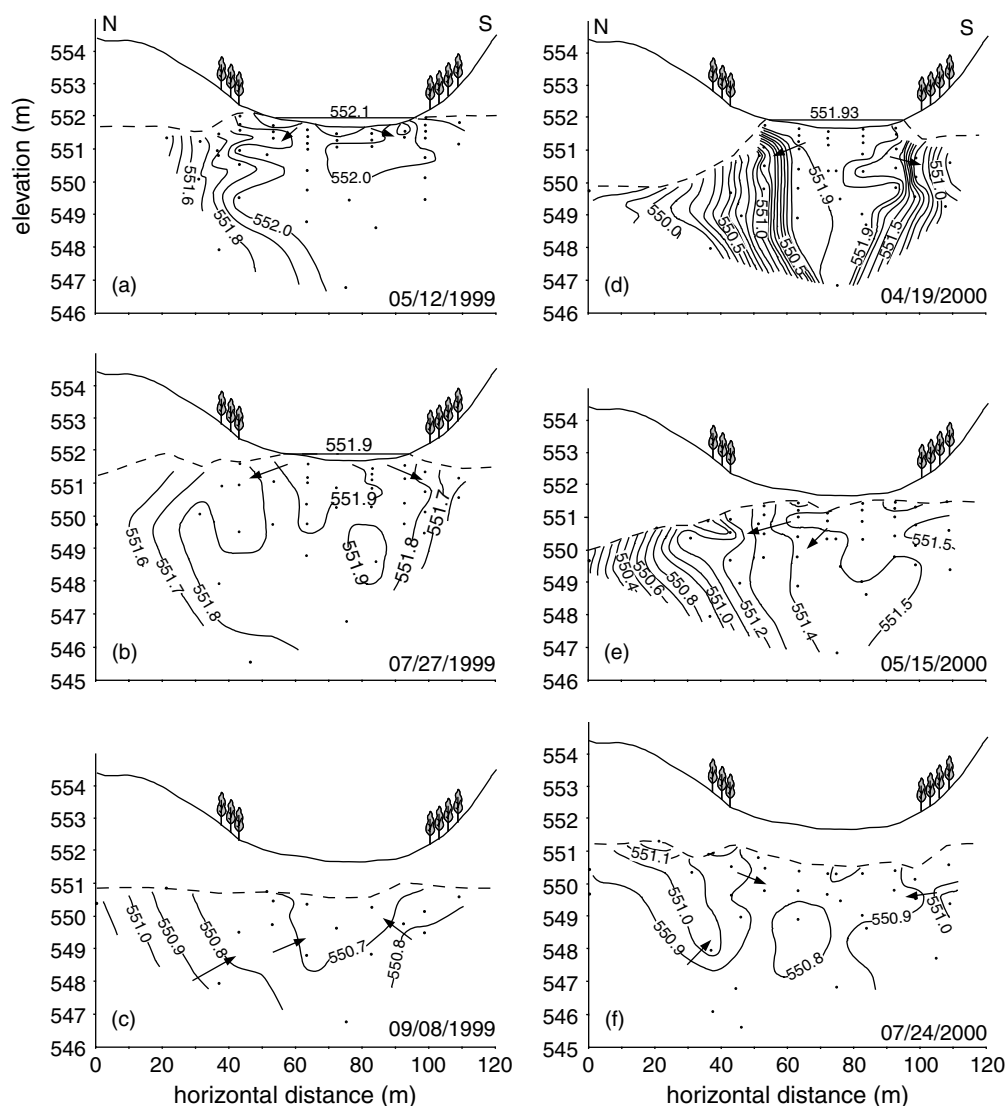


Figure 7. Cross-section of hydraulic head under wetland 109 along N-S in Figure 1. The dashed line indicates the water table, and dots indicate piezometer screens. Arrows show the general pattern of flow directions. The number above the pond indicates the elevation of the pond surface. The date of measurement is indicated on each cross-section

the willow ring (Figure 7b). By late summer, the water table beneath the pond had dropped to below the level of the water table under the upland. This caused the principal flow directions to reverse towards the pond centre from beneath the upland (Figure 7c), in a manner similar to that described by Meyboom (1966).

In the spring of 1999, bromide began to be detected in piezometers, and in some of the deeper ones, bromide was present earlier than expected considering the low hydraulic conductivity of the till (Figure 6). Figure 8 shows sample points and approximate magnitudes of concentration by the size of dots. The distribution of bromide was highly irregular, which could not be meaningfully represented by concentration contours. By mid-May, just two weeks after tracer application, bromide was appearing in piezometers at depths of 2 m in nests 6 and 8 (Figure 8a). This probably was the result of pond water following preferential pathways. This is

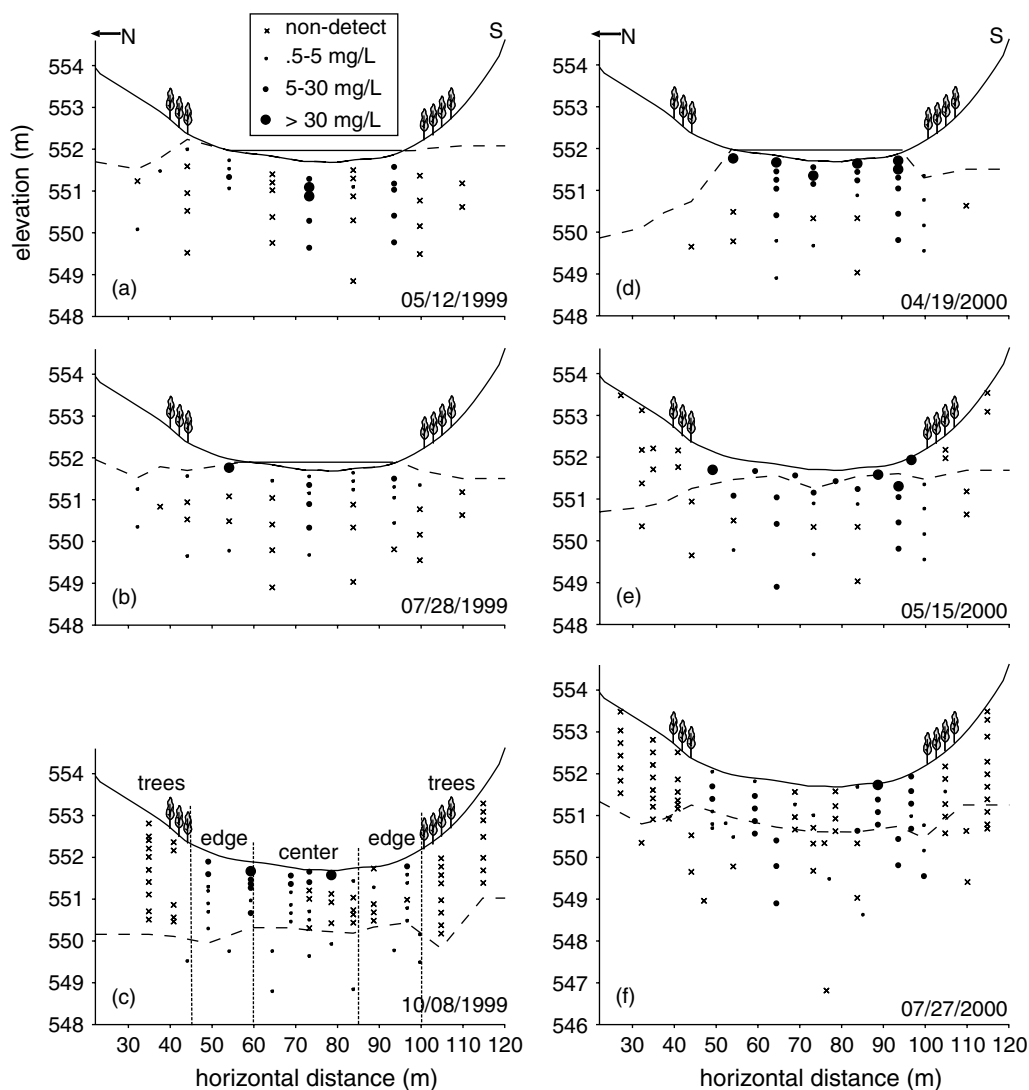


Figure 8. Cross-section of bromide concentration in pore water under wetland 109 along N–S in Fig. 1. The dashed line indicates the water table. The sampling date is indicated on each cross-section. The magnitude of concentration is represented by the size of symbols. All data points in (a), (b) and (d) represent piezometer water samples. In other cross-sections, data points above the water table represent the water samples extracted from the unsaturated soil, and points below the water table represent piezometer water samples

supported by the calculation of the bromide mass balance, presented in the Discussion. Preferential pathways probably are provided by natural features such as fractures in the till or decaying roots. They also could have been conduits formed along the casings of steel piezometers. However, each piezometer in a nest was installed separately, and it is not very likely that all piezometers in a single nest allowed the 'short circuiting' of shallow water. Relatively high concentrations continued to be observed in these particular piezometers throughout the summer (Figure 8b).

As there were uncertainties as to how representative piezometer samples were of subsurface bromide content, a soil sampling survey was conducted on 8–10 October 1999. Figure 8c shows the pore-water bromide concentration of soil samples (above the water table) and piezometer bromide concentration (below the water table). Bromide was mostly concentrated immediately beneath the pond and the pond edges (Figure 8c). Very little bromide was found below 1 m depth beneath the pond centre, and little to no bromide was detected in samples from underneath or outside the willow ring. Some bromide had penetrated to a depth of about 2 m below the pond edges.

In April 2000, the water table was low and steep hydraulic gradients existed between the pond and upland (Figure 7d) because soils adjacent to the pond were still frozen, as confirmed by pushing a frost probe. The pond was dry by early May, and the water table beneath the upland to the south of the pond was at about the same level as the pond, which may have caused a flow-through condition for a brief period (Figure 7e). The water table had declined to 1.5–2.0 m below surface and groundwater flow reversal toward the pond centre had occurred by late July (Figure 7f).

Most bromide was still distributed near the surface in the spring of 2000 (Figure 8d and e). A final set of soil samples collected in July 2000 showed bromide to be more concentrated and reaching deeper levels beneath the pond edges than the centre (Figure 8f). Little to no bromide was detected in soil-water extracts from outside the willow ring.

Vegetation

Chemical analyses of vegetation samples revealed that measurable amounts of bromide had been incorporated into plant tissues of pond vegetation. In a sample of sedge and smartweed collected from the pond centre, 12.7 mg of bromide was detected per gram of dry sample (Table II). Despite the high concentrations, pond

Table II. Bromide content in vegetation samples and the calculation of total bromide uptake by the vegetation

Zone	Location	Vegetation	Br content (mg g ⁻¹)	plant dry mass (g m ⁻¹)	zone area (m ²)	Br mass (kg)
Pond	Centre	Sedge, smartweed	12.7	89	79	0.1
	North	Smartweed	0.08	460	680	0.0
	South	Cow parsnip, sedge, smartweed	9.8	270	680	1.8
	East	Smartweed	0.02	360	680	0.0
	West	Cow parsnip, smartweed, sedge	0.09	330	680	0.0
Trees	North	Willow	0.00	57 100	750	0.0
	South	Aspen, poplar, willow	0.08	3700	750	0.2
	East	Aspen, poplar	0.00	10 400	750	0.0
	West	Willow, poplar	0.00	5200	750	0.0
Upland	North	Grass	0.01	97	400	0.0
	South	Wheat	0.00	370	2090	0.0
	East	Wheat	0.03	470	2090	0.0
	West	Wheat	0.00	700	2090	0.0
Total						2.1

vegetation accounted only for a total bromide mass of about 2.1 kg, as pond plants make up only a very small dry mass (Table II). Trees represented greater plant biomass, but bromide was undetected in three of the four samples from the willow ring. Little to no bromide was found in upland vegetation although some bromide was detected in the sample from the east quadrant of the upland, probably owing to cross-contamination. This possibly occurred during the milling process. Powdered sample had to be cleaned from the mill with a small brush, but it was sometimes difficult to be thorough because of hard-to-access corners, and static electricity. Owing to a small number of samples and the heterogeneity within each zone (Table II), it is difficult to present a meaningful estimate of experimental errors. However, it may be concluded semi-quantitatively that the total vegetation uptake represented a relatively minor portion of bromide in comparison to the total mass released (24 kg).

DISCUSSION

Contributors to pond water-loss

The loss of water from the pond is indicated by the decline of the pond water-level. Meyboom (1966), Millar (1971) and others suggested that the principal mechanism contributing to water loss in ponds is infiltration driven by evapotranspiration at wetland margins. This can be confirmed in wetland 109 by distinguishing between infiltration and evaporation components of pond water-loss and comparing the two.

The infiltration component of water loss can be calculated using the changes in bromide mass in the pond. The bromide mass M is given by the product of concentration C and pond-water volume V . If concentration and water volume are C_1 and V_1 , respectively, at a sampling event, and C_2 and V_2 at the next sampling event, the bromide mass change ΔM between the two sampling events is given by

$$\Delta M = C_2 V_2 - C_1 V_1 \quad (3)$$

Assuming no input of bromide by precipitation or runoff, the mass change should be equal to the amount of infiltration multiplied by concentration

$$\Delta M = -I_{av} \Delta t A_{av} C_{av} \quad (4)$$

where I_{av} , A_{av} and C_{av} are average infiltration rate, pond area and concentration, during the time period, Δt , between the two successive sampling events. From Equations (3) and (4) it follows that

$$I_{av} = \frac{4(C_1 V_1 - C_2 V_2)}{(A_1 + A_2)(C_1 + C_2)\Delta t} \quad (5)$$

where A_1 and A_2 are the pond areas at the times of successive sampling events. The difference between total water loss and infiltration was considered the evaporation from the open water surface. Total water loss was calculated from water-level recession and precipitation.

Figure 9 shows the cumulative infiltration estimate using Equation (5), cumulative total water loss and cumulative open-water evaporation. The infiltration rate was slower in early May 1999, when vegetation is believed to have been less actively transpiring. Linear regression of these data gives a daily infiltration rate of 1.7 mm/day in May and 4.5 mm/day in June–July. Daily open-water evaporation rates were 1.9 mm/day in May and 2.2 mm/day in June–July. Infiltration accounted for 47% of total pond water-loss in May and 67% in June–July. Morton (1983, table A1) reported an open-water evaporation of 3.9–5.1 mm/day in June–July for a large lake in southern Saskatchewan, and Parkhurst *et al.* (1998, figure 13) reported 4.5 mm/day for a larger wetland with few trees around in central North Dakota. These values are much larger than the open-water evaporation in the present study. Using a submerged pan and the Bowen ratio energy balance method, Hayashi *et al.* (1998a) measured an open-water evaporation of 3.0–3.1 mm/day in 29 June to 11 July 1995 in wetland 109, when the potential evaporation on the surrounding upland was expected to be much higher. It

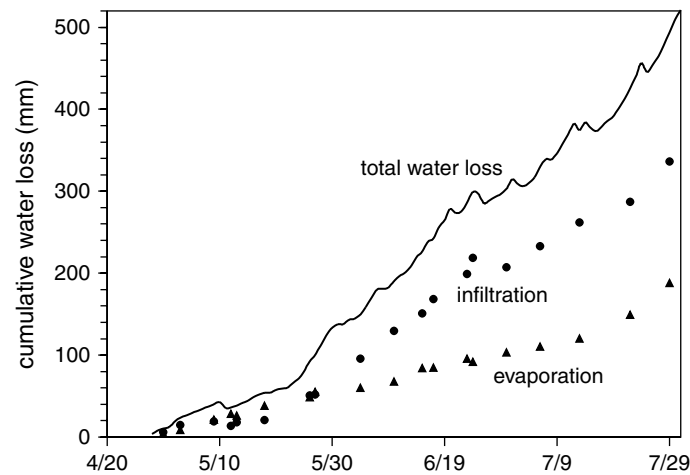


Figure 9. Cumulative water loss (total, infiltration, open-water evaporation) from the central pond in wetland 109 in 1999

is probable that the willow ring blocks wind, thereby reducing the lateral advection of warm and dry air into, and cool and moist air out of, the wetland. Riparian plants intercept the energy advected with warm and dry air and use it, in addition to solar radiation energy, to sustain a high rate of transpiration, thereby inducing the infiltration of water under the pond.

Bromide mass balance

The subsurface distribution of bromide under the wetland was highly irregular (Figure 8), presumably owing to preferential transport pathways. To see the overall pattern of bromide distribution, the cross-section shown in Figure 8c was divided into three zones. The pond centre zone corresponds to the horizontal coordinates of 60–85 m, the pond edge zone corresponds to 45–60 and 85–100 m, and the trees zone corresponds to the remainder. Each zone was further divided into 20-cm-depth intervals, and arithmetic average bromide concentration, C_{av} (kg/m^3), was determined for each depth interval using all samples contained in the interval. Bromide mass content m ($\text{kg Br}/\text{m}^3$ soil) of each depth interval was calculated from

$$m = \theta_{av} C_{av} \quad (6)$$

where θ_{av} is the average volumetric water content estimated from the gravimetric water content and porosity (Figure 5).

Figure 10 shows the bromide mass profiles of the pond centre and edge zones for October 1999. Bromide mass in the trees zone was too small to be shown in Figure 10. The 0–20 cm and 160–180 cm intervals are missing for the edge zone because no sample was analysed from those two depth intervals. Figure 10 indicates that much of the bromide mass is contained between the surface and 40 cm under the pond centre and 80 cm under the edge despite the highly irregular appearance of bromide distribution, as shown in Figure 8c. This suggests that the overall pattern of bromide distribution can be viewed to be undergoing the piston flow condition, although a relatively small amount of bromide is preferentially transported deeper. Figure 9 shows that approximately 300 mm of water infiltrated under the central pond in 1999. Assuming an average porosity of 55–60% (Figure 5), the uniform infiltration front of bromide-laden water would be located at a depth of 50–55 cm, which is roughly consistent with Figure 10. Higher bromide contents in the edge zone suggest that bromide may be concentrated in this zone as a result of the lateral flow and evapotranspiration.

The total subsurface bromide mass was calculated by integrating the depth profile (Figure 10) of each zone, multiplying it by the surface area of the zone, and finally adding all three zones. The estimated total subsurface

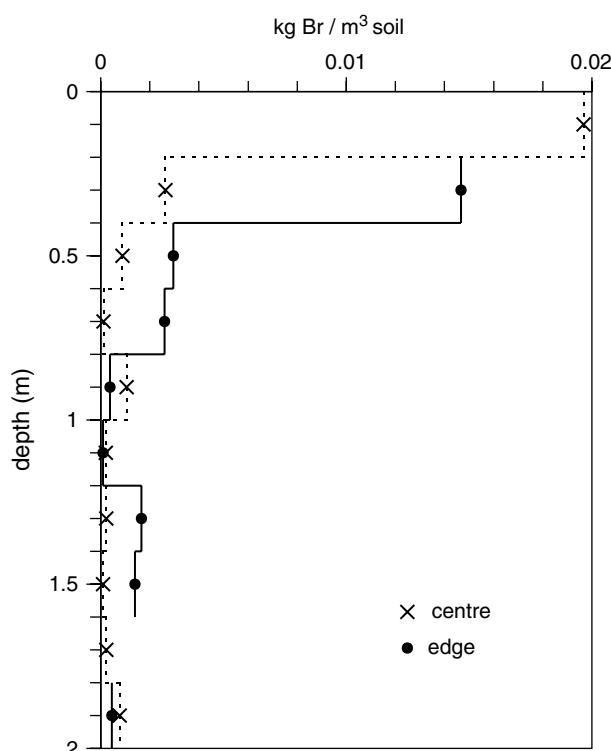


Figure 10. Profiles of average soil bromide content under wetland 109 for the pond centre zone and pond edge zone. The 0–20 cm and 160–180 cm intervals are missing for the edge zone because no sample was analysed from those two depth intervals

bromide in October 1999 was 19.9 kg. Adding to this the vegetation intake of 2.1 kg (Table II) accounts for most of the applied bromide (24 kg). A similar mass-balance calculation was carried out for the data set of 12–13 May 1999, including all piezometer samples. A large amount of bromide (21 kg) was still in the pond water at this early stage of the experiment, and the subsurface mass was expected to be in the order of 3 kg. However, the calculated total subsurface bromide was 11 kg, much higher than expected. Therefore, high concentrations found in piezometers (Figure 8a) must be only representative of a small amount of bromide transported to piezometer screens through discrete macropores.

Recharge of shallow and deep groundwater

High infiltration rates (Figure 9) and the penetration of bromide-laden water (Figures 8 and 10) clearly indicate that wetland 109 is recharging the shallow, local groundwater. However, much of infiltration probably feeds the water uptake by plants, with only a small portion contributing to recharging the larger and deeper aquifer (sand layer at 25 m). A summary of previous studies (Hayashi *et al.*, 1998b) indicates that the wetland-focused recharge, averaged over respective watersheds, is in the order of 1–45 mm/year, which is consistent with published estimates of regional groundwater recharge (5–40 mm/year) in the prairie region (van der Kamp and Hayashi, 1998).

Some water uptake undoubtedly occurred within the pond as indicated by measurable bromide uptake by emergent plants in the pond (Table II). However, the majority of water was probably transpired by the riparian vegetation as suggested by Millar (1971). The hydraulic conductivity of the top 1 m of soil is much higher than material immediately below (Figure 6). It is this high conductivity horizon that probably provided a path along which the lateral flow to the riparian zone occurred. A calculation based on Darcy's law supports

this idea. The horizontal hydraulic gradient in July 1999 was in the order of 0.01 (see Figure 7b). The pond had an approximate area of 750 m^2 and a shoreline length of 100 m in early July. Assuming a hydraulic conductivity of $5 \times 10^{-5} \text{ m/s}$ for the top 1 m (Figure 6), the horizontal flow through this high permeability zone is $5 \times 10^{-7} \text{ m}^3/\text{s}$ per 1 m of the shoreline, and the total flow for the entire shoreline is $5 \times 10^{-5} \text{ m}^3/\text{s}$. If this flow is sustained by the infiltration of pond water, the infiltration rate will be $6.7 \times 10^{-8} \text{ m/s}$ or 5.8 mm/day, which is roughly comparable to the value estimated from the bromide mass balance. This lateral flow persists while the water table under the wetland is above or close to the ground surface (Figure 7b and e), it becomes negligible as the riparian vegetation stops transpiring and the water table becomes flat in winter (Miller *et al.*, 1985; Hayashi *et al.*, 1998a).

CONCLUSIONS

Previous workers have reported that infiltration represents a significant portion of water loss in summer from prairie wetlands. In this study, calculation of a mass balance of a bromide tracer confirms that almost 70% of water level decline in June and July 1999 was from infiltration. This high proportion is largely driven by root uptake of riparian vegetation as shown by the increase in the infiltration component of water loss from early spring to summer as trees and plants begin to grow and transpire more actively. Therefore, much of infiltration only recharged the shallow, local groundwater system and subsequently was lost to evapotranspiration without recharging larger, more regional groundwater systems.

A bromide tracer applied to the pond in April 1999 was found to have accumulated mostly at shallow levels in the underlying soil. The majority of bromide occurred between the soil surface and depths of 40–80 cm in October 1999, suggesting that the transport process associated with the infiltration of pond water into pond bottom sediments can be viewed in the context of the piston-flow model if the bulk behaviour of the dissolved species is concerned. However, a small amount of bromide was found at a depth of 2 m, indicating the presence of preferential flow pathways provided by root holes and fractures.

Vegetation samples obtained in late summer 1999 showed that plants readily incorporated bromide and concentrations became very high within the tissues of individual plants growing within the pond. However, as the total of pond plants amounted to a relatively small dry mass, the total amount of bromide in vegetation is minor in terms of overall mass balance.

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